



ESP8266 OLED Retro Game Console

A mini retro gaming console built using the **ESP8266 (NodeMCU)** and a **0.96" SSD1306 OLED Display**. This project packs multiple classic games, menu navigation, a password unlock system, and bitmap-based graphics into a compact, DIY-friendly handheld device.

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Features



SSD1306 OLED Display

0.96" 128×64 I2C OLED display delivers crisp monochrome visuals with smooth bitmap animations.



13 Built-in Retro Games

A curated collection of classic arcade and puzzle games, all running on a single microcontroller.



Secret Password Unlock

The console boots into a password lock screen — enter the correct button sequence to access the game menu.



5-Button Navigation

UP, DOWN, LEFT, RIGHT, and OK buttons provide intuitive menu navigation and in-game controls.



Score System

Each game tracks your score, adding replay value and a competitive edge to every session.



Retro Arcade Feel

Custom bitmap sprites and smooth animations recreate the authentic feel of classic arcade machines.



Included Game :

#	Game	Description
1	Dino Run	A side-scrolling runner where you jump over obstacles — inspired by the Chrome dinosaur game.
2	Snake	The timeless classic: guide a growing snake to eat food while avoiding walls and yourself.
3	Flappy Stone	Tap to fly through gaps between pipes — a Flappy Bird-inspired challenge with tight controls.
4	Tic Tac Toe	The classic 3×3 grid game, perfect for quick two-player sessions on the go.
5	Pong	The original arcade tennis game — bounce the ball past your opponent's paddle.
6	Mario Mini	A mini platformer featuring Mario-style sprites, pipes, and Goombas on the tiny OLED screen.
7	Car Dodger	Steer your car left and right to dodge oncoming traffic in this fast-paced arcade racer.



Included Games —

#	Game	Description
8	Tappy Tap	A rhythm-based tapping game that tests your timing and reflexes.
9	Space Invaders	Defend against rows of descending alien invaders — a faithful tribute to the 1978 arcade classic.
10	Tetris Lite	Stack falling blocks to clear lines in this simplified but addictive Tetris implementation.
11	Brick Killer	Break all the bricks with a bouncing ball and paddle — an Arkanoid-style breakout game.
12	2048	Slide numbered tiles to combine them and reach the 2048 tile in this popular puzzle game.
13	Doodle Jump	Guide a doodler character upward by hopping on platforms — a vertical jumper with endless replayability.

13

Total Games

All running on a single ESP8266 microcontroller

5

Control Buttons

Simple, intuitive navigation for all games

128

Display Width

128×64 pixel SSD1306 OLED resolution

Made with GAMMA



Hardware Required

This project uses widely available, low-cost components. Everything you need can be sourced from electronics hobbyist suppliers. The total build is beginner-friendly and can be assembled on a breadboard before committing to a PCB.

Core Components

- **ESP8266 NodeMCU** — The main microcontroller board, running the game logic and handling display output via I2C
- **SSD1306 OLED Display (128×64 I2C)** — A 0.96" monochrome display that renders all game graphics and menu screens
- **5 Push Buttons** — UP, DOWN, LEFT, RIGHT, and OK for full navigation and in-game control

Assembly & Power

- **Breadboard / PCB** — Start with a breadboard for prototyping, then migrate to a PCB for a permanent build
- **Jumper Wires** — Standard male-to-male and male-to-female wires for connecting all components
- **Battery (Optional)** — A LiPo or USB power bank can make the console fully portable

Pin Connections

Correct wiring is essential for the console to function. The OLED display uses I2C communication (just two data lines), while each of the five buttons connects to its own GPIO pin on the ESP8266. Follow these pin mappings exactly when assembling your circuit.

OLED Display (SSD1306)

OLED Pin	ESP8266 Pin
VCC	3.3V
GND	GND
SDA	D2
SCL	D1

Control Buttons

Button	GPIO Pin
UP	D6
DOWN	D7
OK	D5
LEFT	D3
RIGHT	D4



Libraries Used

The Adafruit GFX library is the foundational graphics library for many display projects. It provides drawing primitives (lines, circles, rectangles, text) and bitmap rendering support. All game sprites and UI elements in this console are rendered through GFX.

- Core graphics rendering engine
- Bitmap and sprite drawing support
- Font rendering for menus and scores

Adafruit SSD1306

This library provides the hardware-specific interface for the SSD1306 OLED display driver. It handles I2C communication, buffer management, and display refresh. It works in tandem with Adafruit GFX to push rendered frames to the screen.

- I2C display driver interface
- Display buffer management
- Refresh and contrast control

Secret Password Unlock

Default Password Sequence

UP → UP → UP → UP

Press the UP button four times in succession to unlock the console. The sequence is displayed as asterisks or feedback on the OLED screen as you enter it.

Customizable Pattern

The password sequence can be modified in the source code to any pattern of your choosing. This allows you to personalize the unlock sequence or make it more complex for added challenge.

Lock Screen Behavior

The console boots directly into the password screen. Incorrect entries do not unlock the menu — the correct sequence must be entered precisely. Once unlocked, the full game menu becomes accessible.



Graphics & Upload Instructions



Bitmap Graphics

All game sprites are **custom bitmap graphics** designed specifically for monochrome OLED displays. Each sprite is hand-crafted to be recognizable at low resolution while maintaining the retro aesthetic. The sprite library includes:

Dino

Mario

Space Ships

Cars

Goombas

Pipes

Invaders

Bricks

Doodler



How to Upload Code

Follow these steps to flash the firmware onto your ESP8266:

1. **Install Arduino IDE** — Download and install the latest version from arduino.cc
2. **Install ESP8266 Board Package** — Add the ESP8266 board manager URL and install via Tools → Board → Boards Manager
3. **Select Board** — Choose **NodeMCU 1.0 (ESP-12E Module)** from the board menu
4. **Install Libraries** — Install Adafruit GFX and Adafruit SSD1306 via Library Manager
5. **Upload the Code** — Connect the NodeMCU via USB, select the correct COM port, and click Upload



Project Summary

The **ESP8266 OLED Retro Game Console** by **Naresh-sh** is a fully functional, open-source handheld gaming device built on affordable, accessible hardware. It demonstrates what's possible with a \$5 microcontroller and a small OLED display – 13 games, smooth animations, custom graphics, and a password lock system all in one compact build.



Powered by ESP8266

The NodeMCU (ESP-12E) handles all game logic, display rendering, and button input – a testament to the ESP8266's capability beyond WiFi projects.



Open Source on GitHub

The full source code, wiring diagrams, and documentation are available on GitHub under the repository **Naresh-sh/ESP8266-OLED-Retro-Game-Console** – free to fork, modify, and build upon.



Made with ❤️

This project is a labor of love combining retro gaming nostalgia with modern DIY electronics. Perfect for hobbyists, students, and anyone who wants to build their own handheld console.

✅ Ready to build? Head to github.com/Naresh-sh/ESP8266-OLED-Retro-Game-Console for the full README, source code, and commit history. Last updated: May 2026.